

Optimisation report — elios4_ducts_flash_vio

Run

Generated 2026-09-10 16:17 — output folder: exports/optim_smoke/elios4_ducts_flash_vio
Method: random_search budget: 60 evaluations performed: 60 elapsed: 1 s backend: GPU
Decision variables: 21 wall distances: 50 cm, 100 cm, 200 cm eval grid: 30² over 300 cm, 60 rays/pixel

Key results

	initial	best #1	best #2
Score (lower is better)	0.7401	0.4495	0.5000
Uniformity U0 (u0)	58.8 %	60.4 %	55.2 %
FOV coverage	100 %	100 %	100 %
E_avg in FOV (mean over walls)	19,871 lx	15,686 lx	21,167 lx
Active LEDs	24	12	
Drivers	6	3	
Total current	39.6 A	31.7 A	31.1 A
E_min / E_max (last wall)	1,031 / 4,988 lx	838 / 3,904 lx	1,047 / 5,078 lx
ΔEV (stops)	2.27	2.22	2.22
normal: E_avg	2,376 lx	1,756 lx	2,374 lx
normal: VI0 ≥120 lx	47 %	52 %	50 %
flash: E_avg	371,905 lx	185,499 lx	253,843 lx
penalty: beam_angle	0.1528	0.0000	0.0000
penalty: coverage	0.0000	0.0000	0.0000
penalty: driver_cost	0.0600	0.0300	0.0300
penalty: led_cost	0.0480	0.0240	0.0240
penalty: normal.vio	0.0675	0.0000	0.0000

Best design improves the score by 39.3 % over the initial configuration (0.7401 → 0.4495).

The 'initial' column is the optimiser's starting point x₀: the loaded scene when an existing design is refined, or the centre of the search box for a generated layout. The two best designs differ in at least one variable by more than 2 % of its range.

Problem definition

Scene / evaluation settings

Main camera: position (10, -4) cm, pitch 0°, FOV 75° × 60°

Emission: default 168 lm/LED, focus factor 0

Driver model: 6 V, 180 lm/W → 1,080 lm/A, max 13 A per LED, 4 LED(s) per driver

VIO cameras: at [10, 0, 0] cm, cam1 pitch 45° yaw 0°, cam2 pitch -45° yaw 0°, long-side FOV 170° (landscape); VIO wall at 300 cm, 1200 cm wide, 40² cells

Operating modes

- normal: flux × 1; ≥ 50 % of VIO FOV above 120 lx (w=1)
- flash: current 13 A; E_avg ≥ 41,000 lx at 50 cm (w=1)

Objective

metric = u0, E_min percentile = 2 %, coverage weight = 1

Constraints (active)

- max_leds = 24 (weight 0.05)
- led_cost = 0.002
- max_drivers = 6 (weight 0.1)
- driver_cost = 0.01
- max_total_current_a = 40 (weight 1)
- min_led_spacing_cm = 0.8 (weight 1)
- min_beam_angle_deg = 30 (weight 0.5) axis [1, 0, 0]

Decision variables

variable	min	max	int	initial	best #1	best #2
front_duct.n_rows	1	3	●	3	3	3
front_duct.n_cols	1	4	●	4	2	2
front_duct.theta_center	-30	90		30	-1.297	-8.098
front_duct.arc_span	10	100		55	81.91	72.64
front_duct.axial_center	-3	1		-1	-2.321	-3
front_duct.row_pitch	0.8	2		1.4	1.621	1.468
front_duct.radial	-1	1		0	0.1301	0.05167
front_duct.tilt_axial[0]	-60	60		0	38.96	36.37
front_duct.tilt_axial[1]	-60	60		0	-30.48	-15.21
front_duct.tilt_axial[2]	-60	60		0	6.972	3.327
front_duct.tilt_axial[3]	-60	60		0	-37.9	-24.95
front_duct.tilt_axial[4]	-60	60		0	-5.735	7.045
front_duct.tilt_axial[5]	-60	60		0	16.37	8.253
front_duct.tilt_axial[6]	-60	60		0	58.74	60
front_duct.tilt_axial[7]	-60	60		0	60	60
front_duct.tilt_axial[8]	-60	60		0	-43.1	-35.28
front_duct.tilt_axial[9]	-60	60		0	60	60
front_duct.tilt_axial[10]	-60	60		0	44.73	42.93
front_duct.tilt_axial[11]	-60	60		0	-13.26	-28.08
front_duct.beam	90	130		120	92.8	90
front_duct.current_a	0.3	3		1.65	2.641	2.595

Method & formulas

How a design is scored

Each candidate decision vector x is decoded into a regular saved configuration (same JSON schema as configs/*.json), the scene is built with the exact same code as the UI, and the wall illuminance E (lux) is ray-traced on a grid at every wall distance d in the list above. Only the cells inside the main-camera footprint (the FOV trapezoid, drawn in white on the images) are kept; cells with $E = 0$ are ignored for the uniformity metrics but counted for coverage.

Uniformity metrics over the lit FOV cells:

$$E_{min} = P_2(E), \quad E_{avg} = \bar{E}, \quad E_{max} = \max(E)$$

$$U_0 = \frac{E_{min}}{E_{avg}}, \quad U_1 = \frac{E_{min}}{E_{max}}, \quad CV = \frac{\sigma_E}{E_{avg}}$$

E_{min} uses the 2-th percentile of lit cells so a single Monte-Carlo-starved cell cannot dominate the metric.

Score (minimised):

$$S(x) = \frac{1}{N_d} \sum_d [1 - U_0]_d + 1 \left(1 - \min_d \text{cov}_d \right) + \sum_k P_k(x)$$

cov_d is the fraction of FOV cells receiving any light at distance d . Every constraint is a soft penalty $P_k \geq 0$ added to the score (no hard infeasibility), which keeps the landscape usable for population-based global search.

Active penalty terms:

$$P_{led} = 0.002 \cdot N_{LED}$$

$$P_{maxled} = 0.05 \cdot \max(0, N_{LED} - 24)$$

$$P_{drv} = 0.01 \cdot N_{drv}, \quad N_{drv} = \lceil N_{LED}/4 \rceil$$

$$P_{maxdrv} = 0.1 \cdot \max(0, N_{drv} - 6)$$

$$P_l = 1 \max\left(0, \frac{\sum_l I_l - 40}{40}\right), \quad I_l = \frac{\Phi_l}{V \cdot \eta}$$

$$P_{sp} = 1 \sum_{i < j} \max\left(0, 1 - \frac{\|p_i - p_j\|}{0.8}\right)$$

$$P_{ang} = 0.5 \cdot \frac{1}{N} \sum_i \max\left(0, 1 - \frac{\angle(d_i, \hat{a})}{30^\circ}\right), \quad \hat{a} = [1, 0, 0]$$

$$P_{sym} = 0.5 \cdot \frac{\#\{i : \min_j \|p_i - Mp_j\| > 0.5\}}{N}, \quad M = \text{diag}(1, -1, 1)$$

$$\text{mode normal: } E^{(normal)} = sE, \quad s = 1$$

$$P_{normal, vio} = 1 \max\left(0, 1 - \frac{f_{vio}}{0.5}\right), \quad f_{vio} = \frac{\#\{E^{(normal)} \geq 120 \text{ on VIO FOV}\}}{\#\text{VIO FOV cells}}$$

$$\text{mode flash: } E^{(flash)} = sE, \quad s = I_{flash}/\bar{I}$$

$$P_{flash, lux} = 1 \max\left(0, \frac{41000 - E_{avg}^{(flash)}}{41000}\right)$$

Modes share the geometry and differ only by drive current; since luminous flux is linear in current ($\Phi = I \cdot V \cdot \eta$), the traced grid is rescaled per mode instead of re-traced. $\bar{I}$ is the mean design current implied by the LED flux in the decoded configuration.

Optimiser:

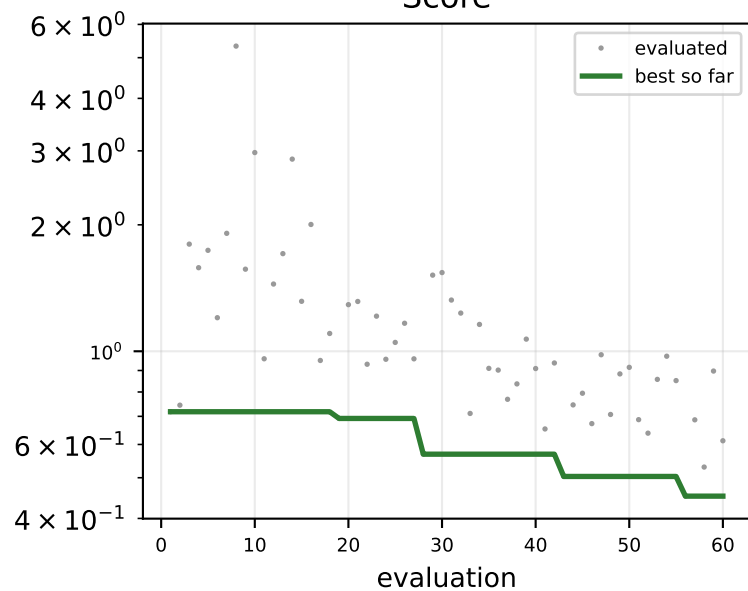
Differential evolution (scipy) samples a population inside the bounds, recombines candidates and keeps improvements; integer variables (on/off states, row/column counts) are rounded before decoding.

Method & formulas

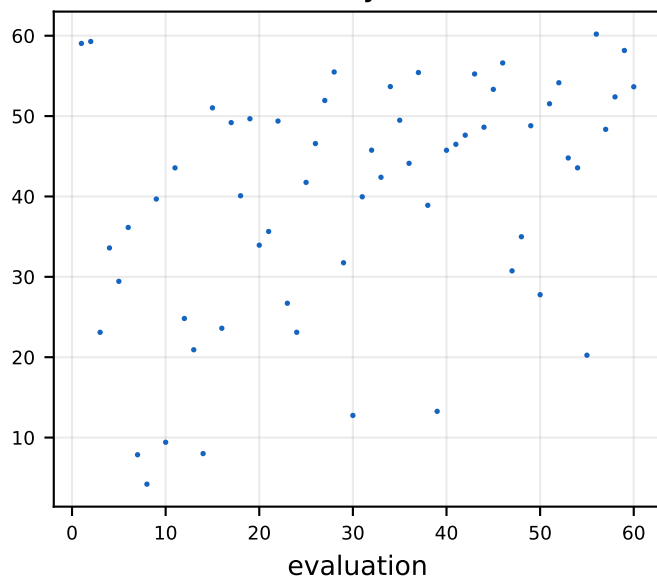
Nelder-Mead and random search are local/simple alternatives used for refinement. The best design is exported after every improvement, so a stopped run is never lost.

Convergence

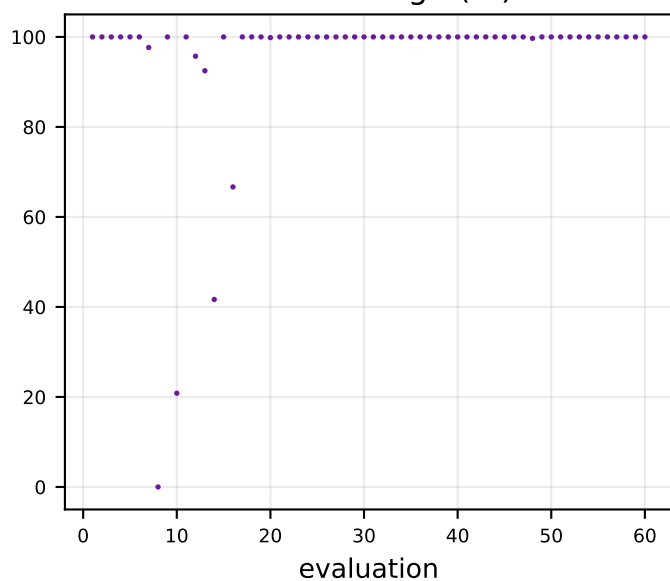
Score



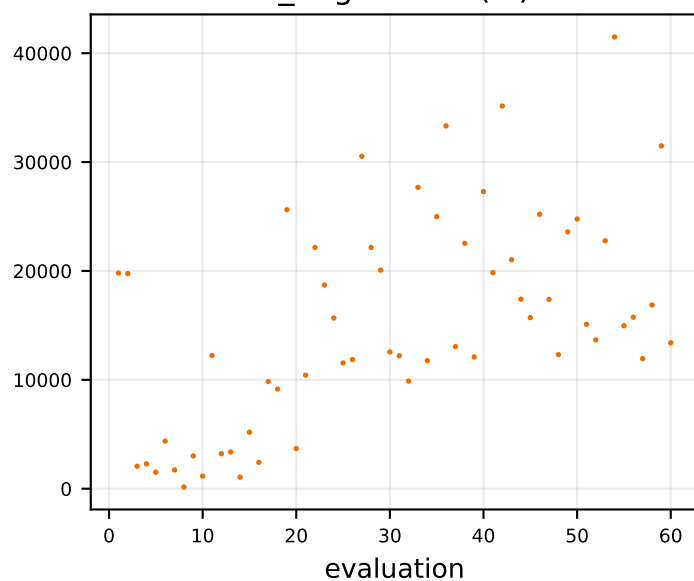
Uniformity U0 (%)



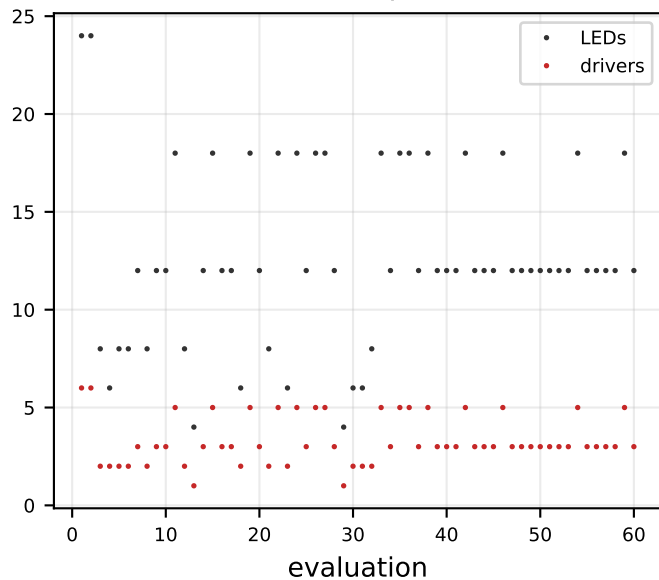
FOV coverage (%)



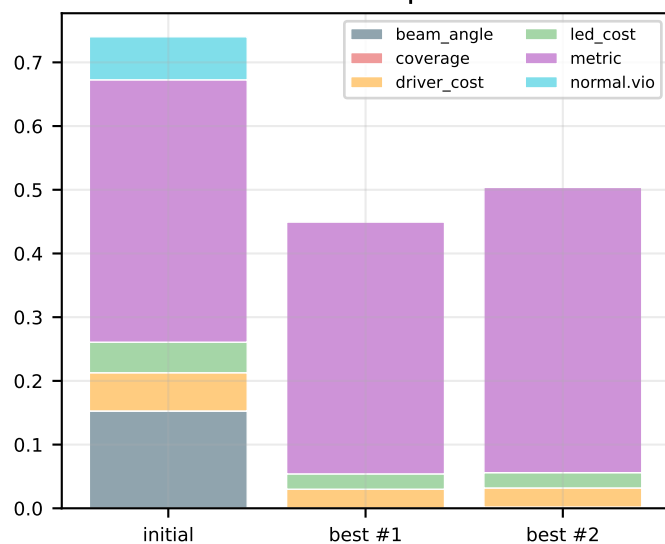
E_avg in FOV (lx)



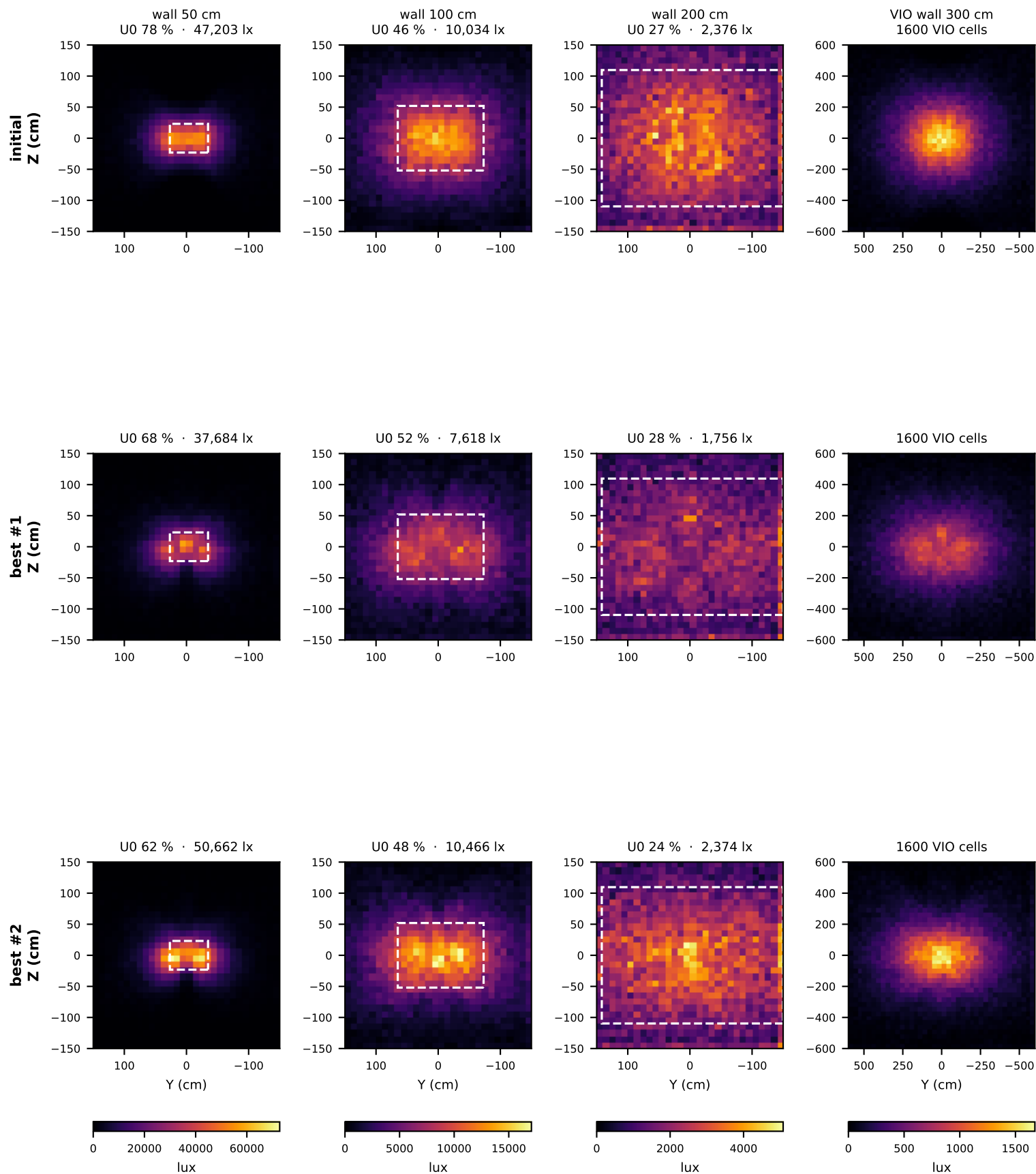
Active LEDs / drivers



Score decomposition

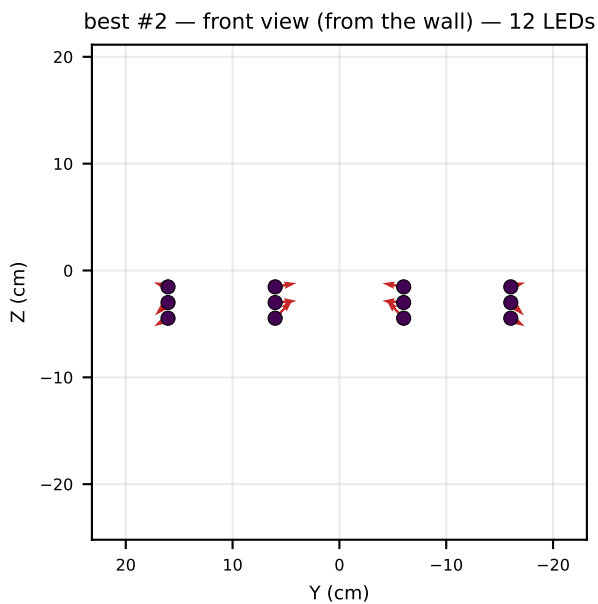
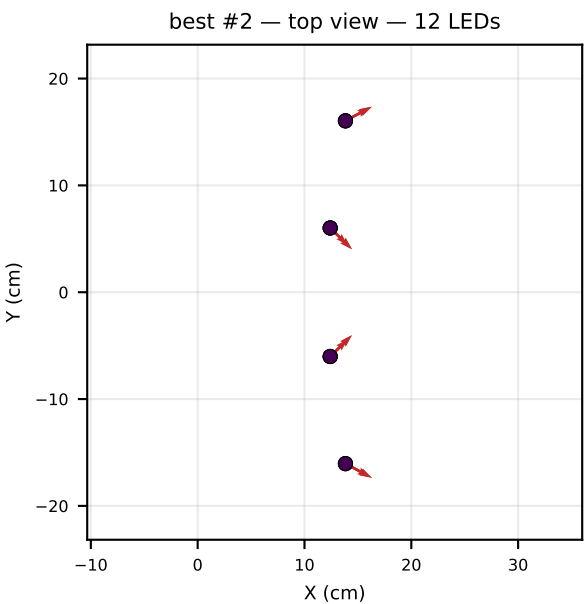
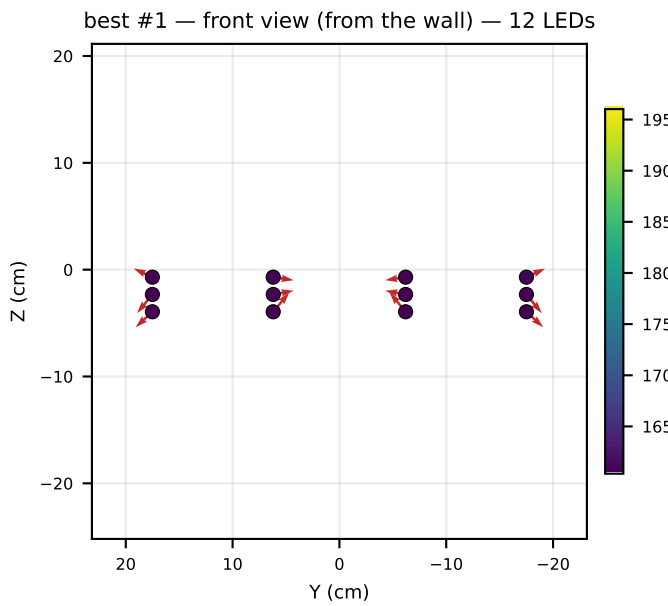
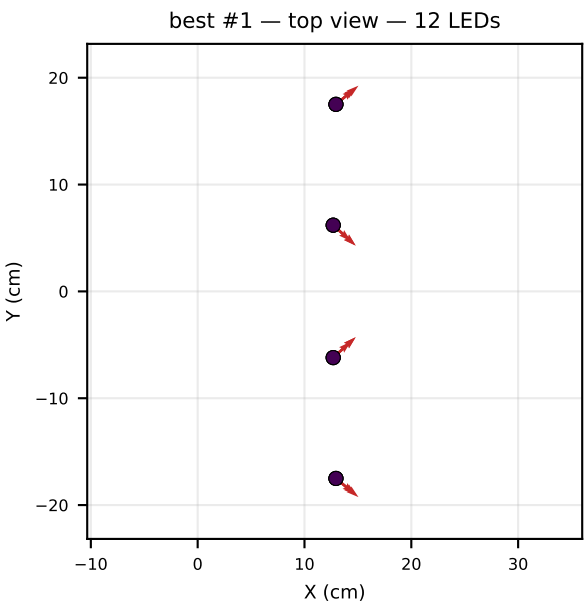
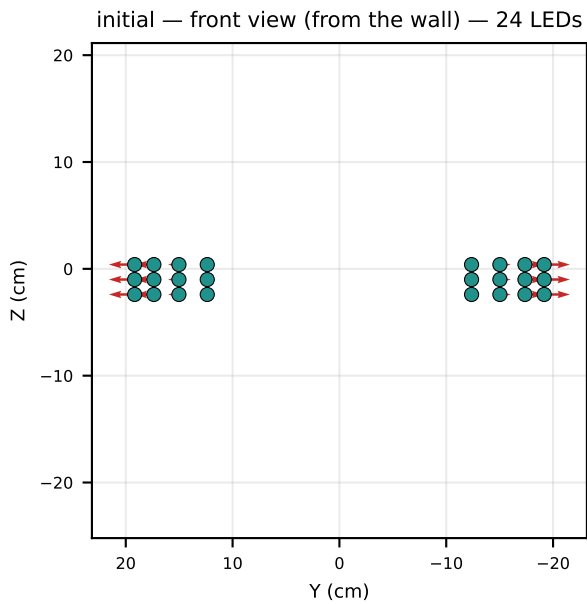
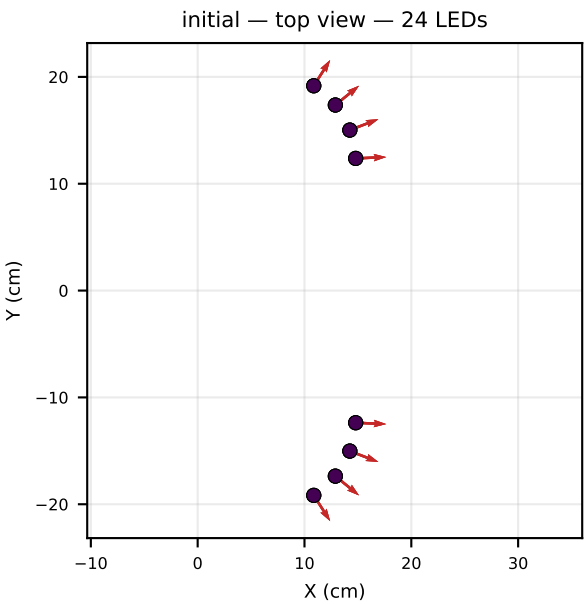


Wall illuminance — initial vs. best designs

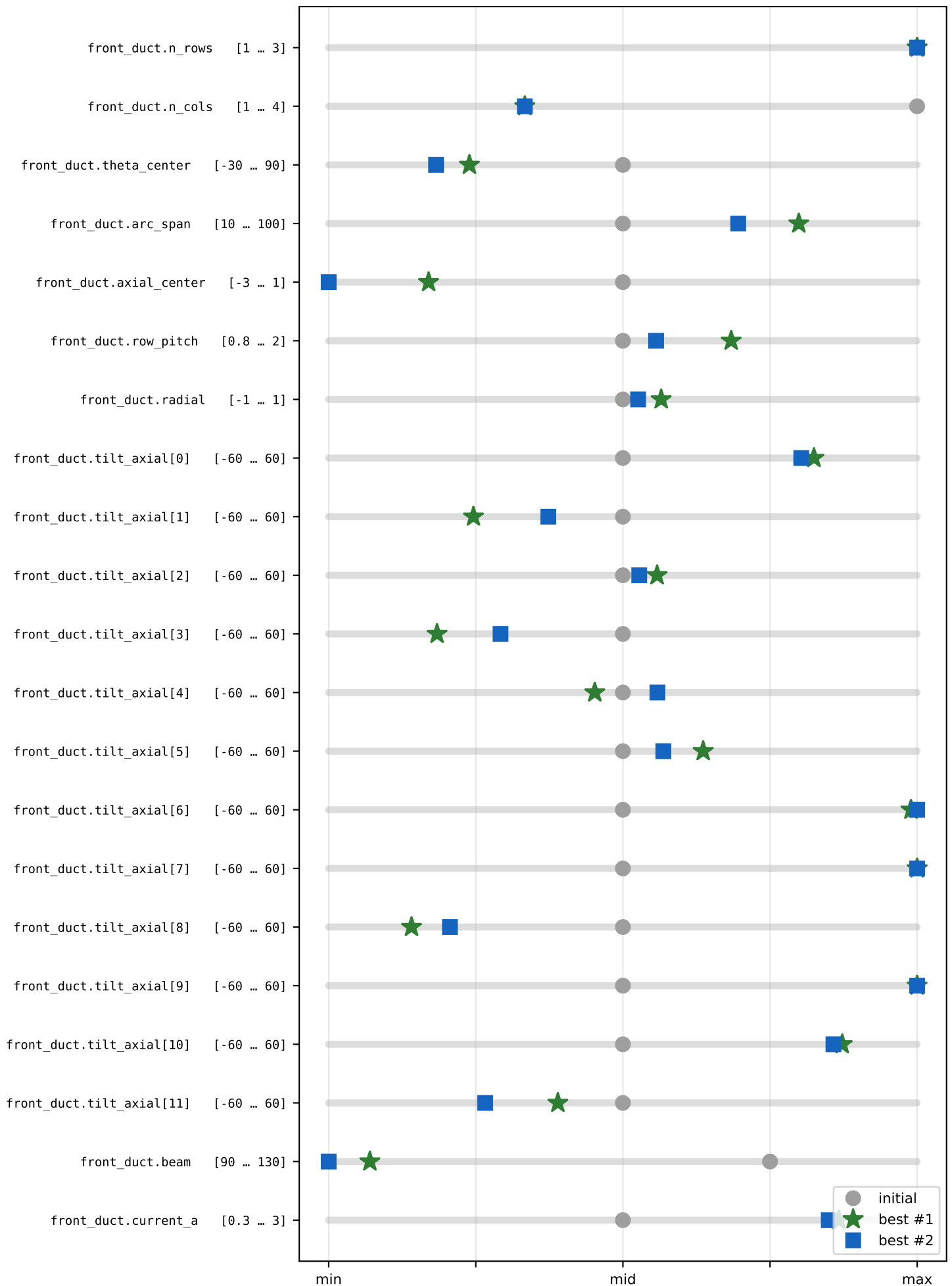


Dashed white: main-camera FOV footprint (metrics are computed inside it). Cyan: VIO camera footprint. Colour scale is shared per column.

LED layouts (active LEDs, arrows = beam axis)

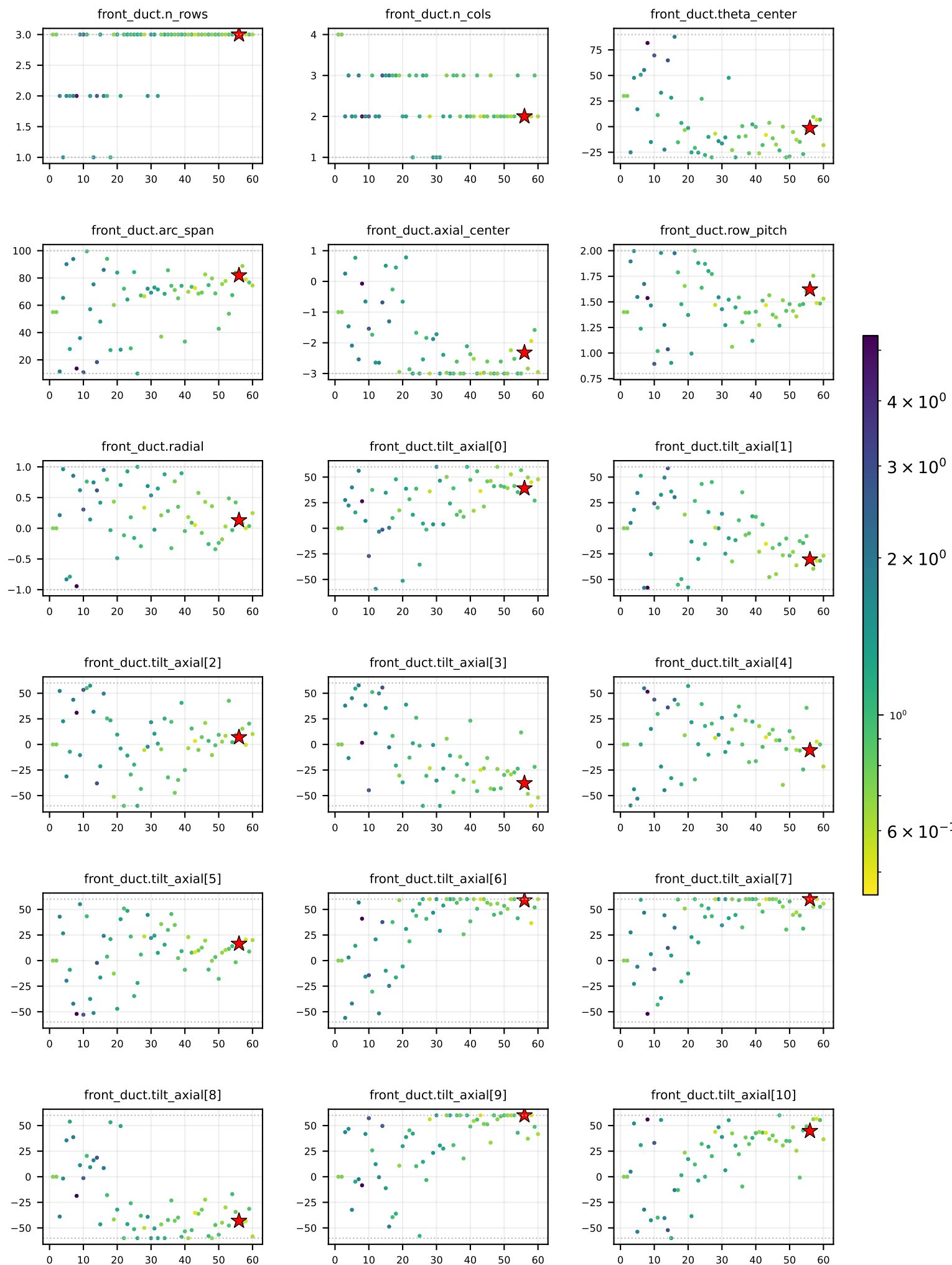


Optimised variables — position inside the search bounds

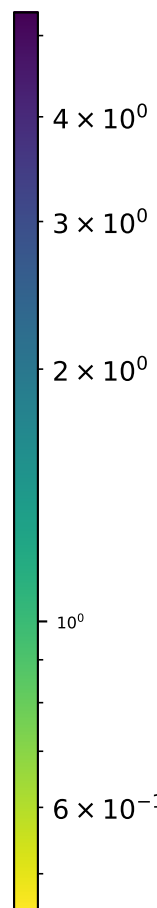
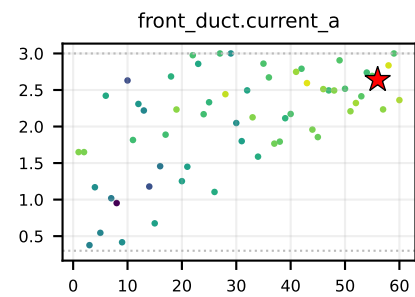
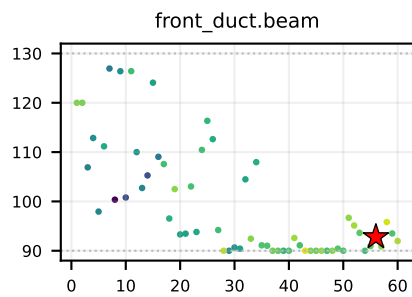
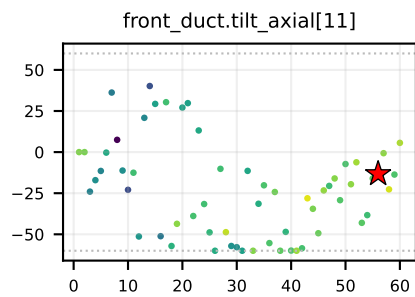


Values sitting on a bound suggest the search box may be too tight for that variable.

Optimised variables — explored values (colour = score, ★ = best)



Optimised variables — explored values (colour = score, ★ = best)



Files

best_config.json — best design, loadable in the UI (Project → Load) or via --evaluate

best2_config.json / initial_config.json — the other two designs compared in this report

history.csv — every logged evaluation with score, metrics and all variable values

summary.json — run metadata, best evaluation and best decision vector

initial: score=0.7401 U=58.8% cov=100% Eavg=19871lx LEDs=24 drv=6 I=39.6A normal[Eavg=2376lx VI0=47%]
flash[Eavg=371905lx] [led_cost=0.048, driver_cost=0.060, beam_angle=0.153, normal.vio=0.068]

best #1: score=0.4495 U=60.4% cov=100% Eavg=15686lx LEDs=12 drv=3 I=31.7A normal[Eavg=1756lx VI0=52%]
flash[Eavg=185499lx] [led_cost=0.024, driver_cost=0.030]

best #2: score=0.5037 U=55.2% cov=100% Eavg=21167lx LEDs=12 drv=3 I=31.1A normal[Eavg=2374lx VI0=50%]
flash[Eavg=253843lx] [led_cost=0.024, driver_cost=0.030, beam_angle=0.002]